ITUPublications
Recommendations

International Telecommunication Union
Standardization Sector

Recommendation

ITU-T G.798 (02/2026)

SERIES G: Transmission systems and media, digital systems and networks

Digital terminal equipments – Other terminal equipment

Characteristics of optical transport network hierarchy equipment functional blocks - Overview

ITU-T G-SERIES RECOMMENDATIONS

Transmission systems and media, digital systems and networks

INTERNATIONAL TELEPHONE CONNECTIONS AND CIRCUITS	G.100-G.199
GENERAL CHARACTERISTICS COMMON TO ALL ANALOGUE CARRIER-TRANSMISSION SYSTEMS	G.200-G.299
INDIVIDUAL CHARACTERISTICS OF INTERNATIONAL CARRIER TELEPHONE SYSTEMS ON METALLIC LINES	G.300-G.399
GENERAL CHARACTERISTICS OF INTERNATIONAL CARRIER TELEPHONE SYSTEMS ON RADIO-RELAY OR SATELLITE LINKS AND INTERCONNECTION WITH METALLIC LINES	G.400-G.449
COORDINATION OF RADIOTELEPHONY AND LINE TELEPHONY	G.450-G.499
TRANSMISSION MEDIA AND OPTICAL SYSTEMS CHARACTERISTICS	G.600-G.699
DIGITAL TERMINAL EQUIPMENTS	G.700-G.799
General	G.700-G.709
Coding of voice and audio signals	G.710-G.729
Principal characteristics of primary multiplex equipment	G.730-G.739
Principal characteristics of second order multiplex equipment	G.740-G.749
Principal characteristics of higher order multiplex equipment	G.750-G.759
Principal characteristics of transcoder and digital multiplication equipment	G.760-G.769
Operations, administration and maintenance features of transmission equipment	G.770-G.779
Principal characteristics of multiplexing equipment for the synchronous digital hierarchy	G.780-G.789
Other terminal equipment	**G.790-G.799**
DIGITAL NETWORKS	G.800-G.899
DIGITAL SECTIONS AND DIGITAL LINE SYSTEM	G.900-G.999
MULTIMEDIA QUALITY OF SERVICE AND PERFORMANCE – GENERIC AND USER-RELATED ASPECTS	G.1000-G.1999
TRANSMISSION MEDIA CHARACTERISTICS	G.6000-G.6999
DATA OVER TRANSPORT – GENERIC ASPECTS	G.7000-G.7999
PACKET OVER TRANSPORT ASPECTS	G.8000-G.8999
ACCESS NETWORKS	G.9000-G.9999

For further details, please refer to the list of ITU-T Recommendations.

Recommendation ITU-T G.798

Characteristics of optical transport network hierarchy equipment functional blocks - Overview

Summary

Recommendation ITU-T G.798 provides an overview of equipment functions that should be used to specify the optical transport network (OTN) functionality of network elements and identifies the Recommendations where these functions are defined.

History *

Edition	Recommendation	Approval	Study Group	Unique ID
1.0	ITU-T G.798	2002-01-06	15	11.1002/1000/5604
1.1	ITU-T G.798 (2002) Amd. 1	2002-06-13	15	11.1002/1000/6057
2.0	ITU-T G.798	2004-06-13	15	11.1002/1000/7329
3.0	ITU-T G.798	2006-12-14	15	11.1002/1000/8983
3.1	ITU-T G.798 (2006) Amd. 1	2008-12-12	15	11.1002/1000/9669
3.2	ITU-T G.798 (2006) Cor. 1	2009-01-13	15	11.1002/1000/9647
4.0	ITU-T G.798	2010-10-22	15	11.1002/1000/10877
4.1	ITU-T G.798 (2010) Cor. 1	2011-04-13	15	11.1002/1000/11117
4.2	ITU-T G.798 (2010) Amd. 1	2011-07-22	15	11.1002/1000/11116
4.3	ITU-T G.798 (2010) Cor. 2	2012-02-13	15	11.1002/1000/11488
4.4	ITU-T G.798 (2010) Amd. 2	2012-04-06	15	11.1002/1000/11487
5.0	ITU-T G.798	2012-12-22	15	11.1002/1000/11778
5.1	ITU-T G.798 (2012) Amd. 1	2014-05-14	15	11.1002/1000/12179
5.2	ITU-T G.798 (2012) Amd. 2	2015-01-13	15	11.1002/1000/12367
5.3	ITU-T G.798 (2012) Cor. 1	2015-08-13	15	11.1002/1000/12529
5.4	ITU-T G.798 (2012) Amd. 3	2017-01-12	15	11.1002/1000/13081
6.0	ITU-T G.798	2017-12-07	15	11.1002/1000/13335
6.1	ITU-T G.798 (2017) Cor. 1	2018-08-06	15	11.1002/1000/13524
6.2	ITU-T G.798 (2017) Amd. 1	2018-08-22	15	11.1002/1000/13518
6.3	ITU-T G.798 (2017) Amd. 2	2019-12-07	15	11.1002/1000/13996
6.4	ITU-T G.798 (2017) Amd. 3	2021-01-13	15	11.1002/1000/14520
6.5	ITU-T G.798 (2017) Cor. 2	2021-05-29	15	11.1002/1000/14621
6.6	ITU-T G.798 (2017) Amd. 4	2022-05-22	15	11.1002/1000/14896
7.0	ITU-T G.798	2023-09-06	15	11.1002/1000/15552
7.1	ITU-T G.798 (2023) Amd. 1	2024-04-22	15	11.1002/1000/15813
7.2	ITU-T G.798 (2023) Amd. 2	2024-12-07	15	11.1002/1000/16069
7.3	ITU-T G.798 (2023) Amd. 3	2025-07-22	15	11.1002/1000/16346
8.0	ITU-T G.798	2026-02-06	15	11.1002/1000/16537

Keywords

Atomic functions, equipment functional blocks, functional specification, optical transport network, OTN.

* To access the Recommendation, type the URL https://handle.itu.int/ in the address field of your web browser, followed by the Recommendation's unique ID.

FOREWORD

The International Telecommunication Union (ITU) is the United Nations specialized agency in the field of telecommunications, and information and communication technologies (ICTs). The ITU Telecommunication Standardization Sector (ITU-T) is a permanent organ of ITU. ITU-T is responsible for studying technical, operating and tariff questions and issuing Recommendations on them with a view to standardizing telecommunications on a worldwide basis.

The World Telecommunication Standardization Assembly (WTSA), which meets every four years, establishes the topics for study by the ITU-T study groups which, in turn, produce Recommendations on these topics.

The approval of ITU-T Recommendations is covered by the procedure laid down in WTSA Resolution 1.

In some areas of information technology which fall within ITU-T's purview, the necessary standards are prepared on a collaborative basis with ISO and IEC.

NOTE

In this Recommendation, the expression "Administration" is used for conciseness to indicate both a telecommunication administration and a recognized operating agency.

Compliance with this Recommendation is voluntary. However, the Recommendation may contain certain mandatory provisions (to ensure, e.g., interoperability or applicability) and compliance with the Recommendation is achieved when all of these mandatory provisions are met. The words "shall" or some other obligatory language such as "must" and the negative equivalents are used to express requirements. The use of such words does not suggest that compliance with the Recommendation is required of any party.

INTELLECTUAL PROPERTY RIGHTS

ITU draws attention to the possibility that the practice or implementation of this Recommendation may involve the use of a claimed Intellectual Property Right. ITU takes no position concerning the evidence, validity or applicability of claimed Intellectual Property Rights, whether asserted by ITU members or others outside of the Recommendation development process.

As of the date of approval of this Recommendation, ITU had received notice of intellectual property, protected by patents/software copyrights, which may be required to implement this Recommendation. However, implementers are cautioned that this may not represent the latest information and are therefore strongly urged to consult the appropriate ITU-T databases available via the ITU-T website at https://www.itu.int/ITU-T/ipr/.

© ITU 2026

All rights reserved. No part of this publication may be reproduced, by any means whatsoever, without the prior written permission of ITU.

Table of Contents

		Page
1	Scope	1
2	References	1
3	Definitions	1
	3.1 Terms defined elsewhere	1
	3.2 Terms defined in this Recommendation	1
4	Abbreviations and acronyms	1
5	Conventions	2
6	Overview of the optical transport network equipment series	2
	6.1 Media element and non-associated overhead	2
	6.2 Optical transmission unit (OTU) and optical data unit (ODU) optical transport network (OTN) layers	3
	6.3 Flexible optical transport network (FlexO)	4

Recommendation ITU-T G.798

Characteristics of optical transport network hierarchy equipment functional blocks - Overview

1 Scope

Recommendation ITU-T G.798 provides an overview of equipment functions that should be used to specify the optical transport network (OTN) functionality of network elements and identifies the Recommendations where these functions are defined.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T G.798.2] Recommendation ITU-T G.798.2 (2026), *Characteristics of optical transport network hierarchy equipment functional blocks – Media.*

[ITU-T G.798.3] Recommendation ITU-T G.798.3 (2026), *Characteristics of optical transport network hierarchy equipment functional blocks - OTU and ODU.*

[ITU-T G.798.4] Recommendation ITU-T G.798.4 (2026), *Characteristics of optical transport network hierarchy equipment functional blocks - FlexO.*

[ITU-T G.872] Recommendation ITU-T G.872 (2024), *Architecture of the optical transport network.*

3 Definitions

3.1 Terms defined elsewhere

None.

3.2 Terms defined in this Recommendation

None.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

AP	Access Point
CP	Connection Point
FlexO	Flexible OTN
FlexOnM	FlexO-n Multiplex section layer
FlexOxR	FlexO-x Regenerator section layer
GCM	Galois Counter Mode
OCh	Optical Channel

OCh-O	Optical Channel – Overhead
ODU	Optical Data Unit
ODUk	Optical Data Unit of level k
ODUP	Optical Data Unit, Path
ODUT	Optical Data Unit, Tandem connection sub-layer
OMS-O	Optical Multiplex Section – Overhead
OSC	Optical Supervisory Channel
OTN	Optical Transport Network
OTS-O	Optical Transmission Section – Overhead
OTSiA	Optical Tributary Signal Assembly
OTSiG	Optical Tributary Signal Group
OTSiG-O	Optical Tributary Signal Group – Overhead
OTU	Optical Transmission Unit
TCMC	Tandem Connection Monitoring Control function
TCP	Termination Connection Point

5 Conventions

None.

6 Overview of the optical transport network equipment series

6.1 Media element and non-associated overhead

Figures 6-1 and 6-2 present the set of atomic functions pertaining to the media element and non-associated overhead of the optical transport network (OTN). The functions and related processes are specified in [ITU-T G.798.2].

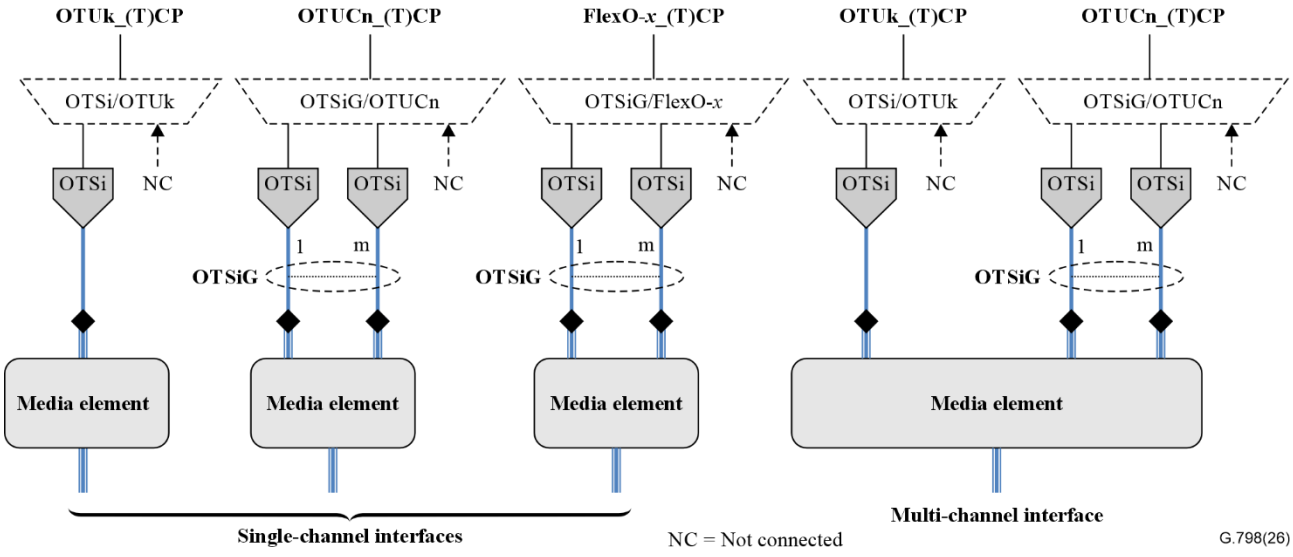


Figure 6-1 – OTN atomic functions specific for the single and multi-channel interface

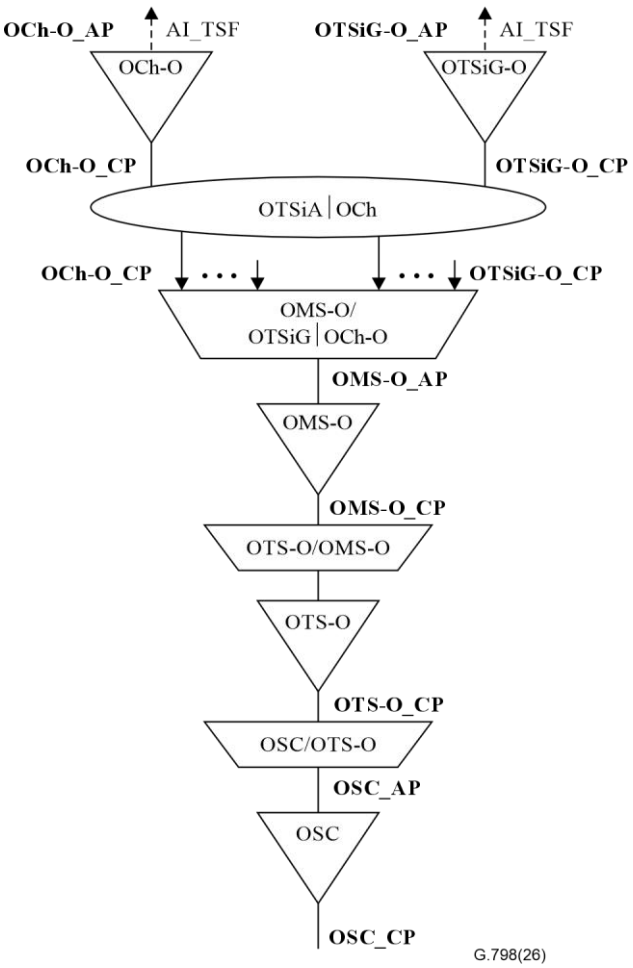


Figure 6-2 – OTN atomic functions specific for the non-associated overhead information

6.2 Optical transmission unit (OTU) and optical data unit (ODU) optical transport network (OTN) layers

Figure 6-3 presents the set of atomic functions pertaining to the OTU and ODU layers, plus adaptation from the media layer to these digital OTN layers. The functions and related processes are specified in [ITU-T G.798.3].

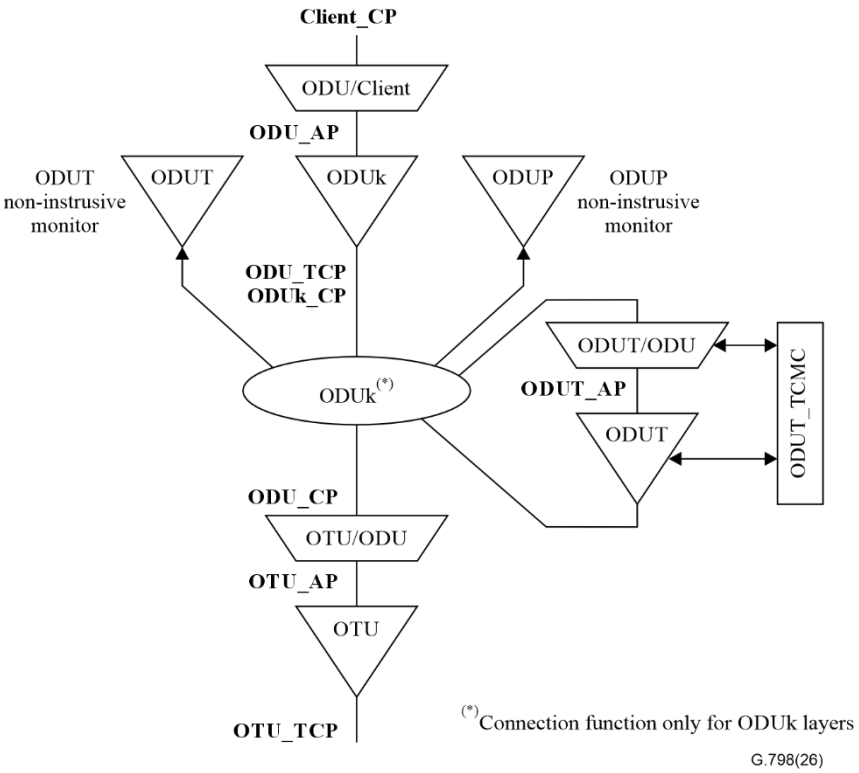


Figure 6-3 – OTU and ODU atomic functions

6.3 Flexible optical transport network (FlexO)

Figures 6-4 and 6-5 present the set of atomic functions pertaining to the FlexO. These functions, as well as the adaptation from the media layer to these layers, and related processes are specified in [ITU-T G.798.4].

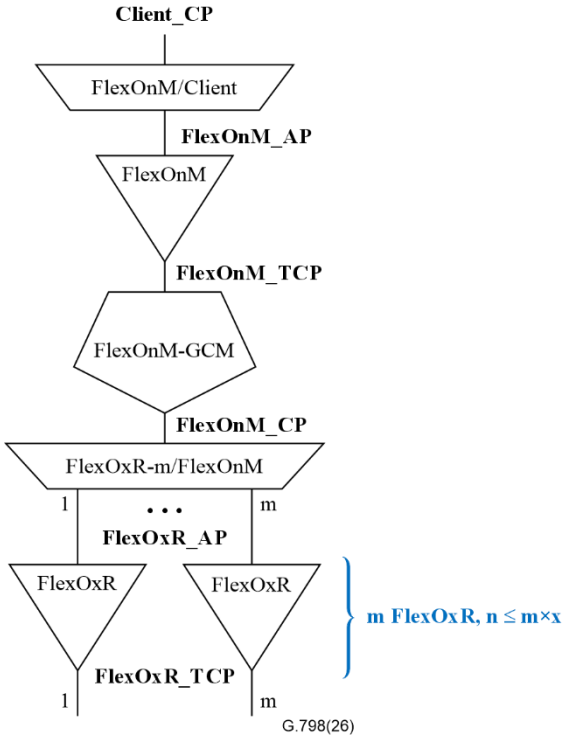


Figure 6-4 – Flexible OTN atomic functions

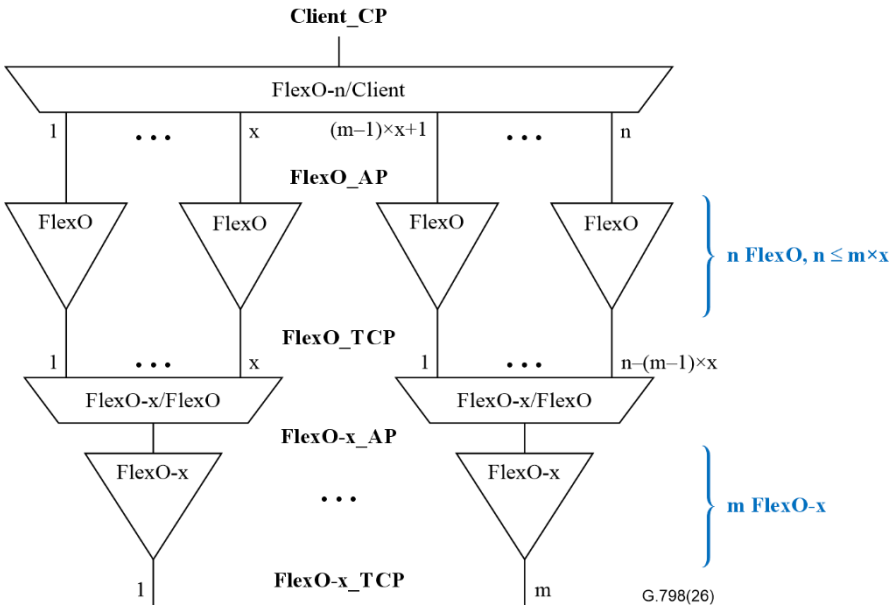


Figure 6-5 – Atomic functions for legacy FlexO model (see Annex A of [ITU-T G.798.4])

SERIES OF ITU-T RECOMMENDATIONS

Series A	Organization of the work of ITU-T
Series D	Tariff and accounting principles and international telecommunication/ICT economic and policy issues
Series E	Overall network operation, telephone service, service operation and human factors
Series F	Non-telephone telecommunication services
Series G	**Transmission systems and media, digital systems and networks**
Series H	Audiovisual and multimedia systems
Series I	Integrated services digital network
Series J	Cable networks and transmission of television, sound programme and other multimedia signals
Series K	Protection against interference
Series L	Environment and ICTs, climate change, e-waste, energy efficiency; construction, installation and protection of cables and other elements of outside plant
Series M	Telecommunication management, including TMN and network maintenance
Series N	Maintenance: international sound programme and television transmission circuits
Series O	Specifications of measuring equipment
Series P	Telephone transmission quality, telephone installations, local line networks
Series Q	Switching and signalling, and associated measurements and tests
Series R	Telegraph transmission
Series S	Telegraph services terminal equipment
Series T	Terminals for telematic services
Series U	Telegraph switching
Series V	Data communication over the telephone network
Series X	Data networks, open system communications and security
Series Y	Global information infrastructure, Internet protocol aspects, next-generation networks, Internet of Things and smart cities
Series Z	Languages and general software aspects for telecommunication systems

Published in Switzerland
Geneva, 2026